



JKBOSE 2026

CLASS 11TH

PERMUTATION AND COMBINATIONS

MATHS

LECTURE-02

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TOPICS TO BE COVERED

1

Exercise Discussion 6.1

2

Factorial Notation

3

Legendre's Formula

4

Exercise Discussion 6.2



Related Problems

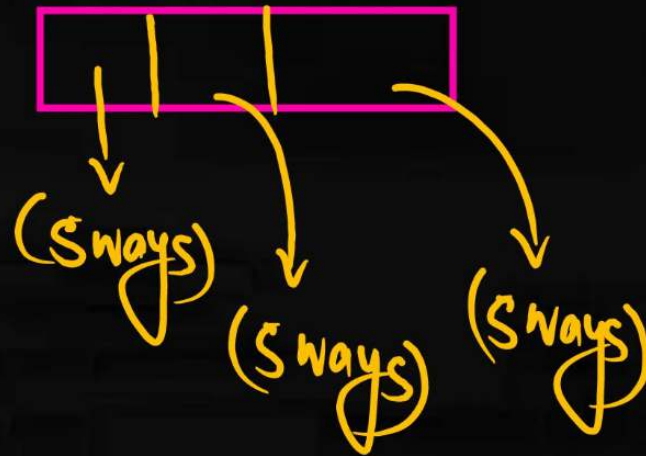
EXERCISE – 6.1

QUESTION

How many 3-digit numbers can be formed from the digits 1, 2, 3, 4 and 5 assuming that

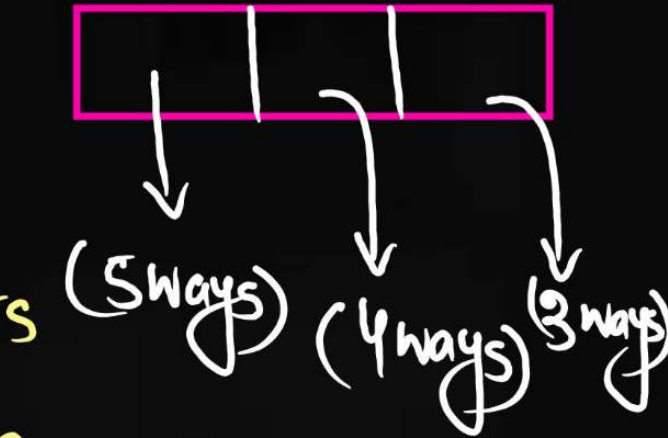
- Repetition of the digits is allowed?
- Repetition of the digits is not allowed?

(Sol) (i) Required no of 3 digit numbers = $5 \times 5 \times 5$
 $= 125$.



(ii)

Required number of 3-digit numbers
 $= 5 \times 4 \times 3 = 60$.

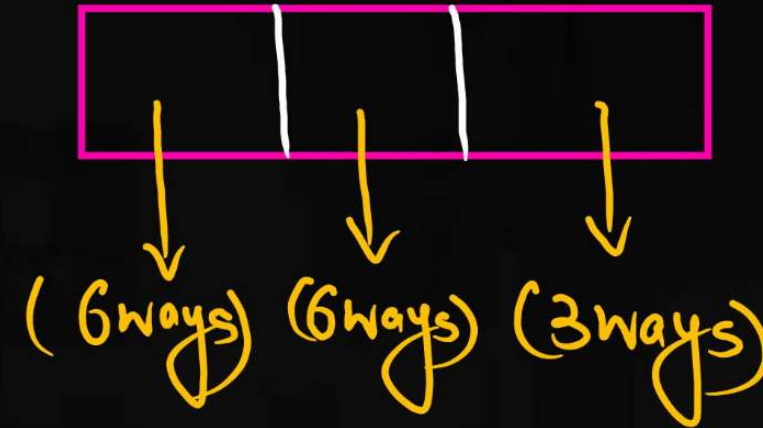


Hence the result.

QUESTION

How many 3-digit even numbers can be formed from the digits 1, 2, 3, 4, 5, 6 if the digits can be repeated?

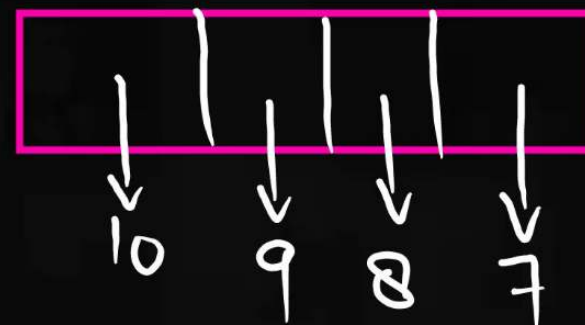
(Sol) Required number of 3 digit
even numbers $= 6 \times 6 \times 3$
 $= 36 \times 3$
 $= 108.$



QUESTION

How many 4-letter code can be formed using the first 10 letters of the English alphabet, if no letter can be repeated?

$$10 \times 9 \times 8 \times 7.$$



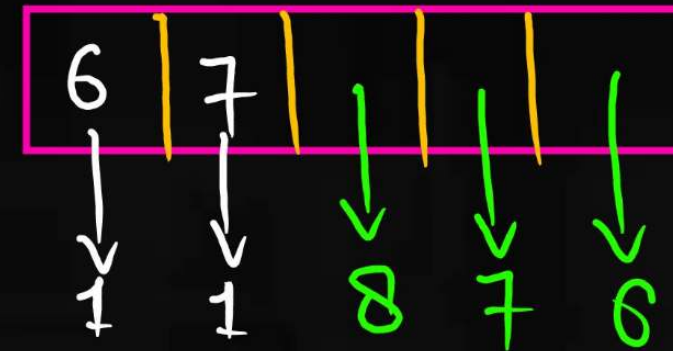
QUESTION

How many 5-digit telephone numbers can be constructed using the digits 0 to 9 if each number starts with 67 and no digit appears more than once?

0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

$$1 \times 1 \times 8 \times 7 \times 6 = ??$$

#



QUESTION

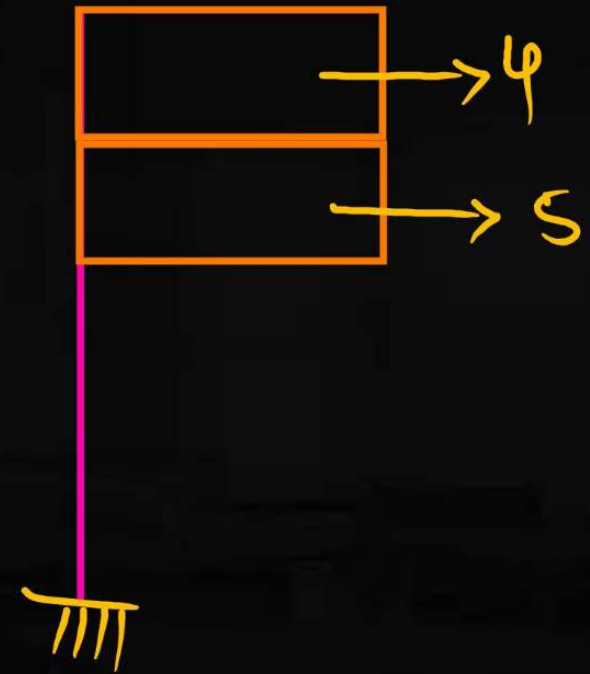
A coin is tossed 3 times and the outcomes are recorded. How many possible outcomes are there?

$$2 \times 2 \times 2 \\ = 8$$

QUESTION

Given 5 flags of different colours, how many different signals can be generated if each signal requires the use of 2 flags one below the other?

(Sol) Required number of signals
 $= 5 \times 4$
 $= 20$. Hence the result



Factorial Notation

✓
 $n!$ or $\underline{1n}$

For any natural number n , $n!$ (n -factorial) represents the product of first n natural number

i.e. $n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

$$\Rightarrow 5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$\Rightarrow 5! = 5 \times 4!$$

$$\Rightarrow 6! = 6 \times 5!$$

$$* n! = n(n-1)!$$

* for $n=0$, we define $0! = 1$

$$* n! = n(n-1)! = n(n-1)(n-2)!$$

EXAMPLE

If $\frac{1}{8!} + \frac{1}{9!} = \frac{x}{10!}$ find x .

(Sol) Given that $\frac{1}{8!} + \frac{1}{9!} = \frac{x}{10!}$

$$\Rightarrow \frac{10!}{8!} + \frac{10!}{9!} = x$$

$$\Rightarrow x = \frac{10 \times 9 \times \cancel{8!}}{\cancel{8!}} + \frac{10 \times 9 \times \cancel{9!}}{\cancel{9!}}$$

$$\Rightarrow x = 90 + 10$$

$$\Rightarrow \boxed{x = 100}$$

EXAMPLE

Find n , if $(n+1)! = 12 \times (n-1)!$

(Sol) Given that $(n+1)! = 12 \times (n-1)!$

$$\Rightarrow (n+1)(\cancel{n+1})(n+1-2)! = 12(n-1)!$$

$$\Rightarrow (n+1)n(\cancel{n-1})! = 12(\cancel{n-1})!$$

$$\Rightarrow (n+1)n = 12$$

$$\Rightarrow n^2 + n - 12 = 0.$$

$$\Rightarrow (n+4)(n-3) = 0$$

$$\Rightarrow n+4 = 0 \text{ or } n-3 = 0$$

$$\Rightarrow n = -4, 3$$

So, $n = 3$.

QUESTION

If $\frac{1}{9!} + \frac{1}{10!} = \frac{n}{11!}$, then the value of n is equal to:

- 1 100 ✗
- 2 110 ✗
- 3 121 ✓
- 4 132 ✗

$$\frac{11!}{9!} + \frac{11!}{10!} = n$$

$$\Rightarrow \frac{11 \times 10 \times \cancel{9!}}{\cancel{9!}} + \frac{11 \times \cancel{10!}}{\cancel{10!}} = n$$

$$\Rightarrow n = 110 + 11$$

$$\Rightarrow n = 121$$

QUESTION

If $(n + 1)! = 90 \times (n - 1)!$, then the value of n is equal to:

1

6X (Sol) Given that $(n+1)! = 90 \times (n-1)!$

2

12X $\Rightarrow (n+1)n \cancel{(n-1)!} = 90 \times \cancel{(n-1)!}$

$\Rightarrow (n+1)n = 90 \rightarrow (n+1)n = 10 \times 9$
 $\Rightarrow n = 9$

3

9✓ $\Rightarrow n^2 + n - 90 = 0$

$\Rightarrow n^2 + 10n - 9n - 90 = 0$

4

10X $\Rightarrow (n+10)(n-9) = 0$

$\Rightarrow n = -10, 9$

$\Rightarrow n = 9$ ✓

QUESTION

If $(n + 2)! = 60 \times (n - 1)!$ then the value of n is equal to:

1

6X

(Sol) Given that $(n+2)! = 60(n-1)!$

2

3✓

$$\Rightarrow (n+2)(n+1)n \cancel{(n-1)!} = 60 \cancel{(n-1)!}$$

$$\Rightarrow n(n+1)(n+2) = 60$$

3

9X

$$\Rightarrow n(n+1)(n+2) = 3 \times 4 \times 5$$

4

7X

So on comparing, we get

$$n = 3.$$

QUESTION

If $\frac{(2x)!}{(2x-2)!} = 90$, then the value of x is equal to:

- 1 $6 \times$ (Sol) Given that $\frac{(2x)!}{(2x-2)!} = 90$
- 2 $3 \times \Rightarrow \frac{2x(2x-1)(2x-2)!}{(2x-2)!} = 90$
- 3 $5 \checkmark \Rightarrow \cancel{2}x(2x-1) = \cancel{90}^{45}$
- 4 $9 \times \Rightarrow x(2x-1) = 45$
 $\Rightarrow 2x^2 - x - 45 = 0$

$$\begin{aligned} &\Rightarrow 2x^2 - 10x + 9x - 45 = 0 \\ &\Rightarrow 2x(x-5) + 9(x-5) = 0 \\ &\Rightarrow (2x+9)(x-5) = 0 \\ &\Rightarrow x = -\frac{9}{2}, 5 \\ &\text{So, } \boxed{x = 5} \end{aligned}$$

QUESTION

Given that $(n + 1)! = 12 \times (n - 1)!$ and $(m + n)! = 720$, then the value of m is :

1

2 ☒

(Sol) Given that $(n+1)! = 12(n-1)!$

$$\Rightarrow (n+1)n(n-1)! = 12(n-1)!$$

$$\Rightarrow n(n+1) = 3 \times 4$$

So, $n = 3$

2

3 ☒

3

4 ☒

Also $(m+n)! = 720 = 6!$

$$\Rightarrow (m+3)! = 6!$$

$$\Rightarrow m+3 = 6$$

$$\Rightarrow m = 3$$

4

6 ☒

EXAMPLE

Prove that $n(n-1)(n-2) \dots (n-r+1) = \frac{n!}{(n-r)!}$

Proof: we have

$$n(n-1)(n-2) \dots (n-r+1) = \frac{n(n-1)(n-2) \dots (n-r+1)(n-r)(n-r-1) \dots 3 \cdot 2 \cdot 1}{(n-r)(n-r-1) \dots 3 \cdot 2 \cdot 1}$$

$$\Rightarrow n(n-1)(n-2) \dots (n-r+1) = \frac{n!}{(n-r)!} \quad \checkmark$$

Hence proved.

$$7 \times 6 \times (7-3+1) = \frac{7!}{(7-3)!}$$

$$\begin{aligned} 7 \times 6 \times 5 \\ = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1} \\ = \frac{7!}{4!} \end{aligned}$$

EXAMPLE

Find the value of t which satisfies $(t - [\sin x])! = 3! 5! 7!$, where $[\cdot]$ denotes the greatest integer

(Sol) we know that

$$-1 \leq \sin x \leq 1$$

$$\Rightarrow 0 \leq |\sin x| \leq 1$$

$$\text{i.e. } |\sin x| \in [0, 1]$$

Case I St: Suppose that $|\sin x| = 1$

$$\text{So } [\sin x] = [1] = 1$$

$$\text{Therefore } (t - [\sin x])! = 3! 5! 7!$$

$$\Rightarrow (t - 1)! = 3! 5! 7! = \overset{\checkmark}{3} \times \overset{\checkmark}{2} \times \overset{\checkmark}{1} \times \overset{\checkmark}{5} \times \overset{\checkmark}{4} \times \overset{\checkmark}{3} \times \overset{\checkmark}{2} \times \overset{\checkmark}{1} \times 7!$$

$$\Rightarrow (t - 1)! = 10 \times 9 \times 8 \times 7!$$

$$\Rightarrow (t - 1)! = 10!$$

$$\text{So } t - 1 = 10$$

$$\Rightarrow t = 11$$

Case 2nd: Suppose that

$$|\sin x| \in [0, 1) \Rightarrow [\sin x] = 0$$

$$\text{Therefore } (t - [\sin x])! = 10!$$

$$\Rightarrow (t - 0)! = 10!$$

$$\Rightarrow t! = 10!$$

$$\Rightarrow t = 10$$

Legendre's Formula

Question: Find the highest exponent of 2 in $4!$. ($E_2(4!)$)

(Sol) we have $4! = 4 \times 3 \times 2 \times 1 = 24$.

$$\Rightarrow 4! = 2^3 \times 3$$

Therefore $E_2(4!) = 3$.

Question: Find the highest exponent of 3 in $6!$. ($E_3(6!)$)

(Sol) we have $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

$$\Rightarrow 6! = 3^2 \times 80$$

$$\begin{array}{r|l} 2 & 24 \\ \hline 2 & 12 \\ \hline 2 & 6 \\ \hline & 3 \end{array}$$

$$\begin{array}{r|l} 3 & 720 \\ \hline 3 & 240 \\ \hline & 80 \end{array}$$

$$\text{So, } E_3(6!) = 2.$$

* Highest exponent of the prime p in $n!$ is given by the formula which is known as Legendre's formula.

$$E_p(n!) = \left[\frac{n}{p} \right] + \left[\frac{n}{p^2} \right] + \left[\frac{n}{p^3} \right] + \dots$$

Where $[.]$ denotes the G.I.F.

Question: Find the highest exponent of 7 in $100!$

(Sol) By Legendre's formula,

$$E_7(100!) = \left[\frac{100}{7} \right] + \left[\frac{100}{7^2} \right] + \left[\frac{100}{7^3} \right] + \dots$$

$$\Rightarrow E_7(100!) = 14 + 2 + 0$$

$$\Rightarrow \boxed{E_7(100!) = 16} \text{ Hence the result.}$$

EXAMPLE

$$2^{47} \mid 50! \quad 2^{22} \mid 50! \quad 6 = 2 \times 3$$

Find the highest exponent of 6 in 50!.

(Sol) By Legendre's formula, we have;

$$E_2(50!) = \left[\frac{50}{2} \right] + \left[\frac{50}{2^2} \right] + \left[\frac{50}{2^3} \right] + \left[\frac{50}{2^4} \right] + \left[\frac{50}{2^5} \right] + \left[\frac{50}{2^6} \right] + \dots$$
$$\Rightarrow E_2(50!) = 25 + 12 + 6 + 3 + 1 + 0$$

$$\Rightarrow E_2(50!) = 47$$

$$\text{Also } E_3(50!) = \left[\frac{50}{3} \right] + \left[\frac{50}{3^2} \right] + \left[\frac{50}{3^3} \right] + \left[\frac{50}{3^4} \right] + \dots$$
$$\Rightarrow E_3(50!) = 16 + 5 + 1 + 0$$

$$50! = 2^{47} \times 3^{22} \times N$$
$$= 2^{22} \times 3^{22} \times 2^{25} N$$
$$= (2 \times 3)^{22} \times 2^{25} N$$

$$\Rightarrow E_3(50!) = 22 \quad \checkmark \quad 6^{22} \times 2^{25} N$$

Therefore the highest exponent of 6 in 50! = $\min(47, 22) = 22$.

Hence the result.

$$\begin{aligned}
 \text{Sol.} &= 2^{47} \times 3^{22} \times N \\
 &= 2^{22} \times 3^{22} \times 2^{25} \times N \\
 &= (2 \times 3)^{22} \times 2^{25} N \\
 &= 6^{22} \times 2^{25} N.
 \end{aligned}$$

QUESTION

The highest exponent of 7 that divides $100!$ is

1 14

2 15

3 16

4 18

H/w.

QUESTION

The highest exponent of 6 that divides $40!$ is

1 18

2 19

3 38

4 20

HP/W

QUESTION

If 3^k is the highest power of 3 that divides $50!$, then the value of k is :

1 22

2 23

3 24

4 25

HP/W

QUESTION

The number of Zeros at the end of $125!$ Is :

1 25

2 30

3 31

4 35

H/W

→ Highest exponent of 10.

$$E_2(125!) ?$$

$$E_5(125!) = ?$$

Homework.

EXERCISE – 6.2

QUESTION



Evaluate

(i) $8!$

(ii) $4! - 3!$

H/w.

QUESTION

Is $3! + 4! = 7!$?

H/W.

QUESTION



Compute $\frac{8!}{6! \times 2!}$

$$\begin{aligned} &= \frac{\overset{4}{\cancel{8}} \times 7 \times \cancel{6}}{\cancel{6} \times \cancel{2}} \\ &= 35 \end{aligned}$$

QUESTION

If $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$, find x

HP/W.

QUESTION



Evaluate $\frac{n!}{(n-r)!}$, when

(i) $n = 6, r = 2 \longrightarrow \frac{6!}{4!} = \frac{6 \times 5 \times \cancel{4!}}{\cancel{4!}} = 30$

(ii) $n = 9, r = 5$

H/w.

$$\begin{array}{r|l}
 2 & 120 \\
 \hline
 2 & 60 \\
 \hline
 2 & 30 \\
 \hline
 3 & 15 \\
 \hline
 & 5
 \end{array}$$

$$5! = 2^3 \times 3 \times 5$$

$$50! = 2^{47} \times 3^{22} \times 5^8$$

Thank
You